Fardin Derogarian Miyandoab – PhD, ASIC Design, R&D, Staff

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Summary

Experienced PhD, Staff ASIC / R&D Engineer with 25+ years of experience spanning semiconductor design, telecommunications, and advanced research. Over 15 years of hands-on expertise in RTL-to-GDSII implementation, physical design, and successful silicon tapeouts across nodes from 350nm to 2nm.

Strong background in hardware/software co-design, embedded systems, and system-level architecture, with additional applied experience in security-relevant hardware behavior, signal integrity, and side-channel–related physical effects (power, timing, EM) through both industrial work and independent experimentation.

Proven ability to analyze complex systems, identify vulnerabilities, optimize architectures, and collaborate across multidisciplinary R&D teams. Highly experienced in automation, scripting, and deep technical problem-solving in advanced semiconductor environments.

Key Expertise

- **Digital IC & SoC Design:** Architecture, RTL, Physical Design, Low Power, RTL-to-GDSII, STA, Synthesis, PnR, CTS, IR/EM, DFT, Tape-out
- **Hardware Security & Robustness (Applied):** Power/timing behavior analysis, side-channel awareness, reliability and variation impacts on silicon
- **High-Speed Interfaces:** DDR, MIPI (CPHY/DPHY), HDMI
- **System & Embedded Design:** Embedded systems, microcontrollers, FPGA prototyping, mixed HW/SW systems
- **EDA & Methodology:** Flow development, automation, design optimization, scripting-driven productivity
- **Advanced Nodes:** TSMC, Samsung, Intel, GF (28nm → 2nm), AMS

Technical Skills

- **EDA Tools:** Cadence (Virtuoso, Encounter, OrCAD), Synopsys (DC, ICC2, Fusion Compiler, DSO.ai, PrimeTime, ICV, StarRC, Formality), Calibre, RedHawk, Altium
- **Scripting:** TCL, Python, Bash
- **Programming:** C, C++, C#, Assembly, Matlab, Database (SQL, MySQL), JS, PHP
- **Hardware Description:** Verilog, VHDL, SystemVerilog
- **Hardware & Prototyping:** FPGA, microcontrollers, embedded systems, PCB design
- **Telecommunications and Networking:** WSN, BSN, IoT, RF, Protocols, Security, Network Architecture

Professional Experience

Synopsys, Porto — Staff ASIC Designer

2018 – Present

- Worked on advanced nodes from 28nm down to 2nm (TSMC, Samsung, Intel, GF).
- Delivered tens of successful IC tape-outs (CPHY, DPHY, MIPI, HDMI, DDR).
- Full backend flow ownership: synthesis → PnR → STA → physical verification → GDSII.
- Performed deep analysis of timing, power, IR/EM, and variation effects, directly relevant to physical behavior and robustness of silicon.
- Developed automation scripts (TCL/Python/shell) improving timing, power, and productivity across teams.
- Collaborated across global teams and mentored engineers in physical design, STA, and automation methodologies.
- Contributed to design robustness and reliability improvements, aligning with security-sensitive silicon behavior considerations.

Instituto de Telecomunicações, Portugal — Postdoctoral Researcher

2015 – 2018

- Research on cognitive radio, routing and MAC protocols for WSN/IoT applications.
- Lectured MSc-level telecommunications courses.
- Authored academic publications and a book.

University of Porto and INESC TEC, Porto — PhD, Researcher

2010 – 2015

- Conducted in-depth study and research on telecommunication networks, with a focus on communication protocols, system-level architectures, and security aspects relevant to embedded and hardware systems.
- Designed a multi-functional router IC for BAN wearable systems: architecture → RTL → PnR → tape-out.
- Fabricated in AMS 0.35µm CMOS.

TCWA Co., Iran — Senior Engineer / R&D Manager

2000 – 2009

- Led R&D projects and managed engineering teams.
- Designed telecom systems (E1-LAN converters, embedded web servers, TMN systems).
- Collaborated with academia and taught engineering courses.

Education

PhD, Telecommunications Engineering — University of Porto, 2015

MSc, Electronic Engineering — University of Urmia, 2003

Patents, Awards & Publications

- Two registered patents in network and communication hardware designs.
- Multiple awards (Best Researcher, Best Paper, Best Engineer).
- Published book and multiple ISI publications (conferences and journals).

References

Available upon request.